Compressing What Matters: Encoder-Aware Visual Token Compression for Taxila Museum Artifact Recognition

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Abstract

Recently, training-free visual token compression has received attention for accelerating vision transformer inference without retraining. However, existing methods are developed and validated almost exclusively on a single supervised backbone, leaving open whether a fixed selection heuristic transfers to encoders trained with different objectives. We show that it does not. Applying ECTS (Early-Contextual Token Selection) across four vision encoders fine-tuned for Taxila Museum artifact recognition, we identify a failure mode we term *selection blindness*: a self-supervised (DINOv2) and a contrastively-trained (SigLIP) encoder each retain substantial removable redundancy under random pruning, yet the same relevance-based scoring criterion consistently picks worse-than-chance subsets, underperforming random removal at every compression level tested (Cohen's d up to ≈ -65). An ablation traces the fault to the scoring function itself, which is actively harmful rather than merely uninformative for these encoders. A representation-drift diagnostic then locates the mechanism: their pruned embeddings shift with 2–7$\times$ higher variance than the supervised backbone's, even when the discarded tokens are spatially indistinguishable from safe ones. Measured latency adds a second correction to the compression story: the selection mechanism itself is slower than the uncompressed model at every retention level except the most aggressive, so token-count savings do not translate to wall-clock savings until pruning is severe. Built on TAD-8K, a curated benchmark of Taxila Museum artifacts extending the recognition system of Khan et al., these results indicate that token redundancy and selection reliability are governed by different properties of an encoder, and that a compression method validated on one backbone cannot be assumed safe on another.

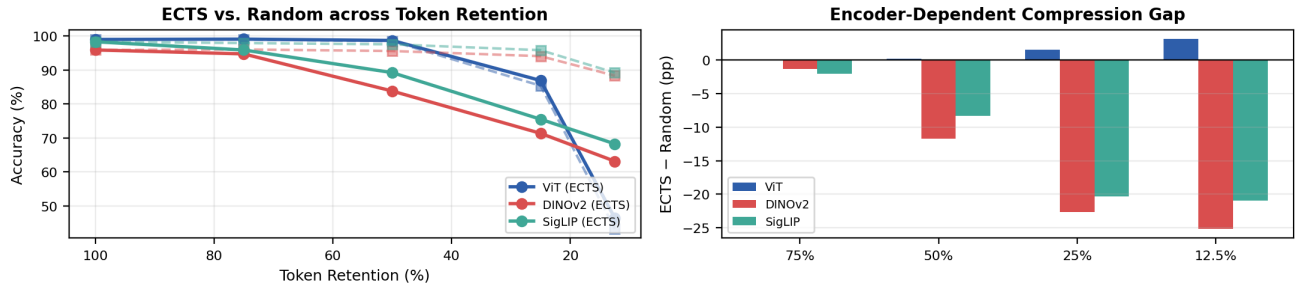


Figure 1: **ECTS is not a universal win.** *(Left)* ECTS (solid) tracks the full-token baseline closely for ViT but falls below Random removal (dashed) for DINOv2 and SigLIP at nearly every retention level. *(Right)* The ECTS−Random accuracy gap is near-zero or positive for ViT but substantially negative for DINOv2/SigLIP, widening as retention drops: the encoder-dependent compression gap that motivates this paper.

1 Introduction

Museums and cultural heritage sites are increasingly adopting AI-assisted visitor experiences, particularly for artifact recognition and guided interpretation on mobile devices. A foundational example of this at the Taxila Museum in Pakistan is Khan et al. [1], who built an augmented-reality smartphone application using a convolutional neural network to recognize artifacts in real time and retrieve supporting multimedia content, and rigorously evaluated its impact on visitor experience against traditional guided and free-form tours. Their work is, to our knowledge, the first to demonstrate a complete, evaluated AI-guided experience specifically for Taxila Museum visitors, and it directly motivates the present study: like Khan et al., we are concerned with artifact recognition *for the Taxila Museum*, and like their system, ours must ultimately answer to the constraint that recognition should run in real time on resource-constrained devices.

This paper continues that Taxila Museum research line, but asks a question specific to the current generation of vision backbones. Since Khan et al.'s system was built, vision transformer (ViT) architectures and their self-supervised (DINOv2 [2]) and contrastive (SigLIP [3]) variants have substantially advanced general-purpose visual recognition, and a substantial literature on visual token compression has emerged to address their computational cost, namely the quadratic scaling of self-attention with token count. Methods such as DynamicViT [4] and EViT [5] learn, end-to-end, which tokens to prune or fuse during training, reporting real measured throughput gains. Training-free alternatives such as Token Merging (ToMe) [19], Adaptive Token Sampling [16], A-ViT [17], SPViT [18], and more recent graph-based methods like Zero-TPrune [21] avoid the retraining requirement but, like their learned counterparts, are evaluated almost exclusively on a single supervised ViT/DeiT-family backbone.

A natural question for a deployment-oriented system is whether a training-free token selection heuristic generalizes not just across images, but across *encoder families with different pretraining objectives*. This is practically relevant: a museum recognition pipeline may be built on any of several backbones (supervised, self-supervised, or contrastively pretrained), and a compression method that only works for one family has limited practical value.

We investigate this question using **TAD-8K**, a curated dataset of 6,930 images across 40 artifact classes from the Taxila Museum, and **ECTS (Early-Contextual Token Selection)**, a training-free token compression method that scores and selects tokens after the first transformer block based on contextual relevance and magnitude, combined with greedy relevance-diversity selection.

Our contributions are fourfold:

- We establish a strong artifact-recognition baseline across four modern vision encoders (ViT-B/16, ConvNeXt-V2 Base, DINOv2-B/14, SigLIP-B/16) on TAD-8K, with ConvNeXt-V2 Base achieving the best accuracy (99.13%).
- We show that ECTS preserves ViT-B/16 accuracy under moderate token compression (within 0.3 percentage points at 50% retention) but *underperforms random token removal* for DINOv2-B/14 and SigLIP-B/16 at every tested retention level, despite both encoders tolerating substantial random token removal, indicating the compression *method*, not the underlying redundancy, is encoder-dependent. This is confirmed with one-sample t-tests against the random-seed distribution.
- A full selection-strategy ablation shows the relevance-and-magnitude scoring criterion is not merely unhelpful but *actively harmful* for DINOv2/SigLIP, and that greedy diversity-aware selection performs substantial, but encoder-dependent and incomplete, corrective work.
- We show, via measured inference latency and a representation-drift diagnostic, that (a) ECTS's token-count reduction does not translate into real speedup under our implementation until very aggressive retention (12.5%), and (b) the ViT/DINOv2/SigLIP divergence is associated with a measurable difference in the *stability* of each encoder's internal representation under pruning, not merely the spatial plausibility of the tokens selected.

We report all of these results, including the negative and cost-related findings, because we believe an honest

account of where and why a training-free compression heuristic fails is more useful to practitioners than a report of where it happens to succeed.

2 Related Work

2.1 AI-Assisted Museum and Cultural Heritage Recognition

Khan et al. [1] proposed an excellent early example of this: an AR-based mobile application for the Taxila Museum that uses a CNN to recognize artifacts in real time and retrieve associated multimedia content, evaluating the system's impact on visitor experience using the Museum Experience Scale and the triptych model of interactivity against traditional tour formats. Their work is the direct predecessor and primary anchor of the present study: both papers study artifact recognition specifically for the Taxila Museum, and both are motivated by the same real-time, resource-constrained deployment setting.

A growing body of recent work (2024–2026) applies modern deep learning to museum and cultural-heritage recognition more broadly. Ipalakova et al. [7] compared eight architectures, including ConvNeXt-Tiny and Swin-based models, for museum exhibit recognition from video-derived datasets. Wen and Ma [8] and Xu et al. [9] both propose deep learning integration for museum information/retrieval systems, the latter reporting 94.3% accuracy with real-time retrieval. Several works focus specifically on Egyptian cultural heritage: Esmat et al. [10] classify artifact materials with lightweight architectures; Hesham et al. [11] propose an on-device, retrieval-augmented recognition system for the Grand Egyptian Museum; Abdelfattah and Atef [12] and Abdelhady et al. [13] combine AR and machine learning for Egyptian heritage guides. Liu et al. [14] and Alhazmi et al. [15] extend similar recognition pipelines to broader cultural-artifact and landmark identification tasks.

Most recently, Huang et al. [6] proposed MAMS-Net, a multimodal system for museum artifact classification and management that combines YOLOv9-based artifact localization, dual-branch visual representation learning with ConvNeXt-V2 and DINOv2, cross-attention fusion of visual and textual metadata, and a management-oriented indexing head for retrieval and inventory consistency. MAMS-Net is considerably broader in scope than the present work: it targets an end-to-end recognition-to-management pipeline incorporating textual metadata and retrieval, whereas we are confined to the narrower question of visual recognition accuracy and token-level compression behavior within individual vision encoders. Notably, two of our four compared encoders (ConvNeXt-V2, DINOv2) overlap with MAMS-Net's visual backbones; our work can be read as a complementary, encoder-focused analysis of exactly the kind of visual representations such management-oriented systems would need to compress or deploy efficiently.

Our work differs from all of the above in scope: rather than building or evaluating an end-user AR system or a multimodal management pipeline, we focus on the underlying recognition backbone, comparing modern vision encoder architectures on a purpose-built Taxila artifact dataset (TAD-8K), and investigating whether these more capable but more computationally expensive backbones can be compressed for deployment in a manner analogous to the resource constraints Khan et al.'s system targets.

2.2 Vision Transformer Token Compression

A substantial body of work reduces the computational cost of vision transformers by pruning, merging, or otherwise reducing the number of tokens processed by later layers. DynamicViT [4] inserts lightweight prediction modules trained end-to-end to progressively prune tokens, reporting 31–37% FLOP reduction and over 40% throughput improvement with under 0.5% accuracy loss. EViT [5] reorganizes tokens during the forward pass guided by class-token attention, fusing rather than discarding less-attentive tokens. Both are trained jointly with the backbone and evaluated primarily on a single encoder family.

A parallel line of work targets training-free compression, closer in spirit to ECTS. Adaptive Token Sampling [16] and A-ViT [17] adaptively determine token budgets per image; SPViT [18] performs latency-aware soft pruning; Token Merging (ToMe) [19] merges rather than discards similar tokens without any retraining; Heo et al. [20] and Zero-TPrune [21] both propose fully training-free pruning using attention-graph or activation

statistics; Bian et al. [22], Lin et al. [23], Xu et al. [24], Zhu [25], and Kim et al. [26] (ToFu, a hybrid pruning-merging approach) explore related directions; see the recent survey by [27] for a broader overview of this rapidly growing area. Closer to our efficiency analysis, Heo et al. [28] and the AdaMerge method [29] both report *measured* latency for training-free acceleration, a practice we follow but find does not straightforwardly hold for our own method. Most relevant to our representation-drift analysis, Lu et al. [30] study anomalous, high-activation "artifact tokens" and attention sinks that emerge within vision transformers, and Venkatraman et al. [31] identify a related failure mode in self-evolving large multimodal models, *visual under-conditioning*, where generation relies on language priors rather than sufficient attention to visual tokens, addressed via invariance-based rewards rather than token selection. Both offer plausible structural accounts for why some encoders' or models' token representations are more fragile or unreliable under manipulation than others, echoing our own representation-drift findings from a different angle (training-time regularization vs. post-hoc compression).

Our work differs in two respects. First, ECTS is training-free and applied post hoc to an already-fine-tuned classifier, without any auxiliary pruning objective. Second, and more central to our contribution, we evaluate compression not on a single encoder family but across three architecturally and objective-wise distinct encoders (supervised ViT, self-supervised DINOv2, and contrastively-trained SigLIP), allowing us to observe that a fixed, training-free selection heuristic does not transfer across these objectives, a question the largely single-backbone prior literature does not directly address. We further report measured latency and find that, unlike several of the learned and training-free methods above, our selection mechanism does not realize comparable real-world speedup except at very aggressive retention, because the selection cost itself is non-trivial without a lightweight learned predictor.

Before arriving at ECTS, we implemented and rejected two simpler training-free candidates as controls: a magnitude-based Top-K selector and a redundancy-aware candidate (ECRAS) combining magnitude with cosine diversity, both of which underperformed random token removal substantially (Section 3), reinforcing that naive magnitude- or redundancy-based heuristics are inadequate for this task and motivating ECTS's post-first-block contextual scoring.

2.3 Encoder Pretraining Objectives and Representation Structure

The three transformer backbones compared in this work represent three distinct pretraining paradigms. ViT-B/16 [32], as used here, is fine-tuned from ImageNet-21k supervised pretraining. DINOv2 [2] is trained with a self-distillation and masked-image-modeling objective on a large curated unlabeled dataset, producing general-purpose features intended to transfer without fine-tuning across tasks and domains. SigLIP [3] replaces CLIP's softmax-normalized contrastive loss with a per-pair sigmoid loss, enabling stable training at large batch sizes and producing image-text aligned representations. These differing objectives are a plausible source of the encoder-dependent compression behavior we observe: a supervised classification objective may concentrate task-relevant information into a readily identifiable subset of tokens (favorable to a relevance-based heuristic like ECTS), while self-supervised and contrastive objectives, optimized for general-purpose transfer rather than a single downstream label, may distribute discriminative information more diffusely or less predictably across tokens. We provide direct evidence consistent with this hypothesis via a representation-drift analysis (Section 5) rather than relying on this pretraining-objective argument alone.

3 Methods

3.1 Dataset

We use TAD-8K, a dataset of artifact photographs collected at the Taxila Museum, spanning 40 artifact classes. From an initial collection of 8,068 images, 1,138 exact duplicate files were removed. A near-duplicate analysis identified 8,302 near-duplicate pairs across 1,205 clusters involving 4,492 images; these clusters were used to construct a group-aware train/validation/test split, ensuring no near-duplicate image group crosses a split boundary. Table 1 summarizes the final dataset. All images were resized to 224×224 during preprocessing.

Table 1: TAD-8K dataset statistics.

Statistic	Value
Original images collected	8,068
Exact duplicates removed	1,138
Final cleaned dataset	6,930
Number of artifact classes	40
Near-duplicate pairs identified	8,302
Images involved in near-duplicate clusters	4,492
Near-duplicate clusters	1,205
Train split (group-aware)	4,849
Validation split (group-aware)	1,045
Test split (group-aware)	1,036
Classes missing from test set	0
Image resolution (preprocessing)	224 x 224

3.2 Vision Encoders

Four pretrained vision encoders were fine-tuned on TAD-8K for 40-class artifact classification: ViT-B/16 [32] (196 spatial tokens, 14×14 patches), ConvNeXt-V2 Base [33] (a convolutional architecture, included in the recognition comparison but excluded from the token compression analysis as it has no discrete token sequence or self-attention mechanism to which ECTS applies), DINOv2-B/14 [2] (256 spatial tokens, 16×16 patches, self-supervised pretraining), and SigLIP-B/16 [3] (196 spatial tokens, 14×14 patches, contrastive image-text pretraining). All models were implemented via `timm` (1.0.24) on PyTorch 2.5.1 (CUDA 12.1), trained and evaluated on a single NVIDIA RTX 3050 (8GB).

3.3 Baseline Evaluation Protocol

Test-set predictions were saved for each fine-tuned model. To rule out stale or inconsistent inference artifacts, we performed a manual forward-pass equivalence check for each transformer encoder, comparing saved predictions against freshly computed inference and comparing manual (block-by-block) forward computation against the library's built-in forward pass; all comparisons showed zero prediction difference and zero maximum logit difference.

3.4 Prior Candidate Methods (Rejected)

Before arriving at ECTS, two simpler training-free selection strategies were implemented and evaluated as controls. A magnitude-based Top-K baseline (retaining tokens with the highest raw patch-embedding L2 norm) performed substantially *worse* than random token removal at every retention level for all three encoders (e.g., at 50% retention: ViT 96.04% vs. random 98.42%; DINOv2 63.13% vs. random 95.54%; SigLIP 68.73% vs. random 97.51%). A subsequent candidate, ECRAS (Encoder-Calibrated Redundancy-Aware Token Selection), combined magnitude with cosine-similarity-based redundancy scoring and adaptive weighting, but likewise underperformed random removal substantially (e.g., DINOv2 at 25% retention: 35.71% vs. random 94.03%). Both results motivated the contextual, post-first-block scoring approach used in ECTS.

3.5 ECTS: Early-Contextual Token Selection

ECTS is a training-free token compression method applied to already-fine-tuned transformer classifiers, with no retraining step (Figure 2). Given an input image: (1) patch and positional embeddings are computed as in the base architecture; (2) the first transformer block is applied to the full token sequence, providing early

contextual information; (3) prefix and spatial tokens are separated, and a per-token relevance-and-magnitude score is computed; (4) tokens are selected via a greedy relevance-diversity procedure; (5) the selected spatial tokens are reattached to the prefix tokens and passed through the remaining transformer blocks and classification head unchanged.

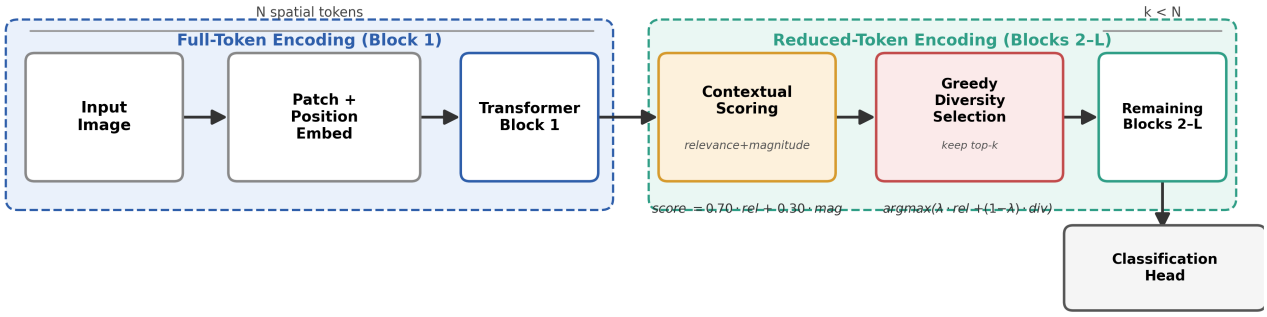


Figure 2: **ECTS pipeline.** Tokens are contextualized by the first transformer block before scoring, then a subset of $k < N$ tokens is selected via greedy relevance-diversity selection and processed by the remaining blocks. Only the selected tokens incur the cost of the remaining transformer layers.

Scoring. For ViT and DINOv2, relevance is the cosine similarity between each spatial token and the (now contextualized) CLS token; for SigLIP, which has no CLS token, relevance is cosine similarity to the mean-pooled spatial token as a global prototype:

$$\text{relevance}_i = \frac{x_i \cdot c}{\|x_i\| \, \|c\|} \tag{1}$$

where x_i is the i-th spatial token (post block 1) and c is the contextual reference. Normalized magnitude is

$$\text{magnitude}_i = \frac{\|x_i\|}{\text{mean}_j(\|x_j\|)} \tag{2}$$

and the combined score is

$$\text{score}_i = 0.70 \cdot \text{relevance}_i + 0.30 \cdot \text{magnitude}_i. \tag{3}$$

Selection. Tokens are chosen greedily to maximize relevance and diversity jointly. The adaptive trade-off parameter is

$$\lambda = \text{clamp}\big(0.25 + 0.50 \cdot \bar{s},\ 0.25,\ 0.75\big) \tag{4}$$

where $\bar{s}$ is the mean pairwise cosine similarity across all spatial tokens in the image. For a candidate token i and already-selected set S, diversity is

$$\text{diversity}_i = 1 - \max_{j \in S} \frac{x_i \cdot x_j}{\|x_i\| \, \|x_j\|} \tag{5}$$

and the final greedy selection score is

$$\text{final}_i = \lambda \cdot \text{relevance}_i + (1 - \lambda) \cdot \text{diversity}_i. \tag{6}$$

Tokens are added one at a time, always selecting $\arg\max(\text{final}_i)$, until the retention budget $k = \text{round}(N \cdot r)$ is met, for retention ratio $r \in \{1.00, 0.75, 0.50, 0.25, 0.125\}$ and full token count N (196 for ViT/SigLIP, 256 for DINOv2).

3.6 Random Token Removal Baseline

As a control, we evaluate random token removal at the same four retention levels using five fixed random seeds (42, 123, 456, 789, 1001), reporting mean and standard deviation across seeds.

3.7 Selection-Strategy Ablation

To isolate the contribution of each ECTS component, we additionally evaluate, at 50% retention on the full test set for all three transformer encoders: relevance-only scoring, magnitude-only scoring, relevance+magnitude scoring without diversity-aware selection, and the full ECTS method.

3.8 Statistical Testing

We compute a one-sample t-test using the reported random-seed mean and standard deviation ($n = 5$) as the reference distribution and the ECTS accuracy as the test value:

$$t = \frac{\text{ECTS} - \mu_R}{\sigma_R/\sqrt{n}}, \quad d = \frac{\text{ECTS} - \mu_R}{\sigma_R} \tag{7}$$

evaluated against a t-distribution with $n - 1 = 4$ degrees of freedom, with d as Cohen's effect size. This is a one-sample test against summary statistics rather than a two-sample test on raw per-seed values, since only mean/std were retained from the random-seed runs.

3.9 Inference Latency

Real per-image inference latency was measured on the RTX 3050 at batch size 1, with 10 untimed warm-up iterations followed by 200 timed iterations per (encoder, retention) configuration, using `torch.cuda.synchronize()` immediately before and after each timed forward pass. We report speedup relative to each encoder's own full-token baseline:

$$\text{speedup} = \frac{\text{latency}_{100\%}}{\text{latency}_r}. \tag{8}$$

3.10 Representation-Drift Diagnostic

For each of 300 test images, we measured the cosine similarity and relative L2 distance between the final pooled embedding (immediately prior to the classification head) from a full-token forward pass (e_{full}) and an ECTS-pruned forward pass (e_{pruned}) of the same image, at both 50% and 25% retention:

$$\cos_\text{sim} = \frac{e_{\text{full}} \cdot e_{\text{pruned}}}{\|e_{\text{full}}\| \, \|e_{\text{pruned}}\|}, \quad \text{drift} = \frac{\|e_{\text{full}} - e_{\text{pruned}}\|_2}{\|e_{\text{full}}\|_2}. \tag{9}$$

3.11 Qualitative Token Selection Visualization

To complement the quantitative drift analysis, we visualized the spatial locations of tokens selected by ECTS and by random removal at 50% retention for a sample of test images across all three transformer encoders, overlaying selected/discarded patches on the original image.

3.12 Theoretical Attention-Compute Reduction

Self-attention cost scales quadratically with token count. If N is the full token count and k the retained count, the pairwise attention compute fraction is $(k/N)^2$, i.e. a reduction of $\left(1 - (k/N)^2\right) \times 100\%$. This describes only the pairwise attention term in layers *after* pruning, excluding the unpruned first block, the selection overhead itself, and non-attention components; hence Section 4 reports *measured* latency rather than relying on this formula alone.

4 Results

4.1 Artifact Recognition Baseline

Four vision encoders were fine-tuned and evaluated on the held-out TAD-8K test set (Table 2, Figure 3). ConvNeXt-V2 Base achieved the highest test accuracy (99.13%, Macro-F1 98.77%), narrowly ahead of ViT-B/16 (98.94%, 98.52%) and SigLIP-B/16 (98.26%, 97.95%). DINOv2-B/14 trailed the other three by a noticeably larger margin (95.85%, 95.62%). Per-class results (Appendix B) show only 8 of 40 classes fall below perfect F1, driven by small, easily-confused classes.

Table 2: **Artifact recognition baseline comparison** across four vision encoders on the TAD-8K test set (1,036 images, 40 classes). Best result in **bold**.

Model	Accuracy (%)	Macro-F1 (%)
ConvNeXt-V2 Base	**99.13**	**98.77**
ViT-B/16	98.94	98.52
SigLIP-B/16	98.26	97.95
DINOv2-B/14	95.85	95.62

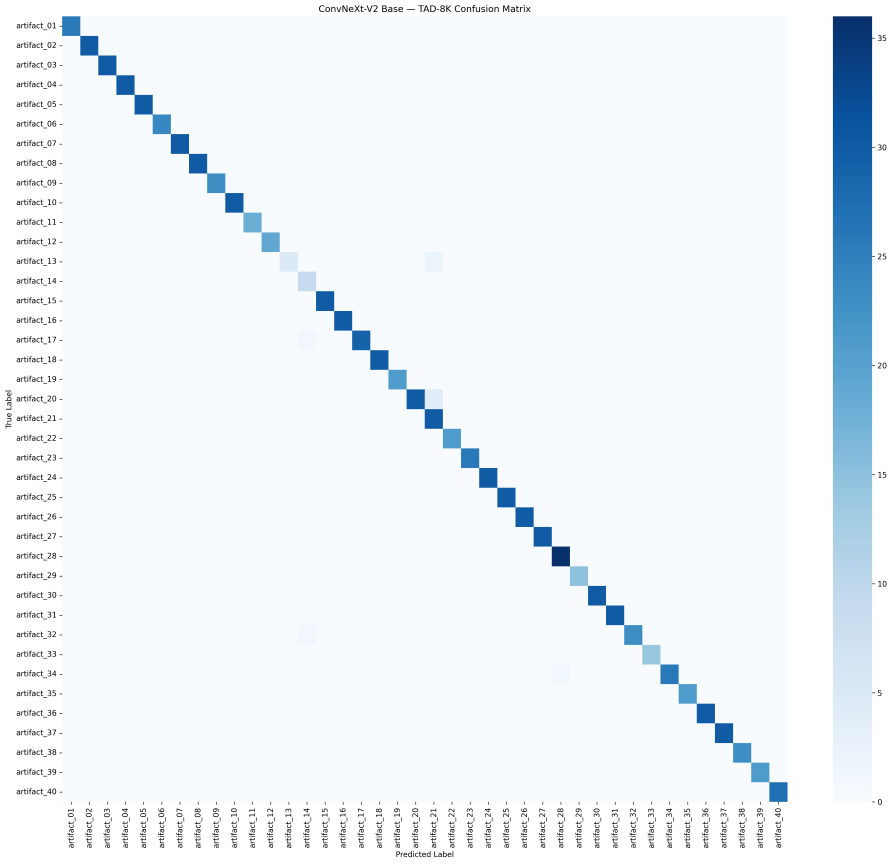


Figure 3: Confusion matrix for ConvNeXt-V2 Base (best-performing encoder) on the TAD-8K test set.

4.2 Encoder-Aware Token Compression (ECTS)

Table 3 and Figure 4 report accuracy and Macro-F1 for Baseline, Random, and ECTS across four retention levels. For ViT-B/16, ECTS closely tracks the full-token baseline at 75% and 50% retention and modestly outperforms random removal at every retention level. For DINOv2-B/14 and SigLIP-B/16, both encoders tolerate random token removal remarkably well (DINOv2 retains 94.03% accuracy even at 25% retention under random removal), yet ECTS underperforms random removal at every retention level for both encoders, by as much as 11.8pp (DINOv2, 50%) and 8.3pp (SigLIP, 50%). Table 4 confirms these differences are statistically significant with large effect sizes for the failure cases.

Table 3: **Baseline vs. Random vs. ECTS** across four retention levels, three transformer encoders. Random values are mean over 5 seeds. **ECTS** rows in bold.

Model	Retention	Tokens	Method	Acc. (%)	F1 (%)	Acc. Δ (pp)	F1 Δ (pp)
ViT	100.0%	196	Baseline	98.94	98.52	0.00	0.00
ViT	**75.0%**	**147**	**ECTS**	**99.03**	**98.73**	**-0.10**	**-0.20**
ViT	75.0%	147	Random	98.90	98.49	0.04	0.03
ViT	**50.0%**	**98**	**ECTS**	**98.65**	**98.34**	**0.29**	**0.19**
ViT	50.0%	98	Random	98.42	98.07	0.52	0.45
ViT	**25.0%**	**49**	**ECTS**	**86.87**	**86.32**	**12.07**	**12.21**
ViT	25.0%	49	Random	85.33	84.53	13.61	13.99
ViT	**12.5%**	**24**	**ECTS**	**46.43**	**44.12**	**52.51**	**54.41**
ViT	12.5%	24	Random	43.30	41.13	55.64	57.40
DINOv2	100.0%	256	Baseline	95.85	95.62	0.00	0.00
DINOv2	**75.0%**	**192**	**ECTS**	**94.69**	**94.26**	**1.16**	**1.36**
DINOv2	75.0%	192	Random	96.00	95.73	-0.15	-0.11
DINOv2	**50.0%**	**128**	**ECTS**	**83.78**	**82.63**	**12.07**	**12.99**
DINOv2	50.0%	128	Random	95.54	95.30	0.31	0.32
DINOv2	**25.0%**	**64**	**ECTS**	**71.33**	**68.39**	**24.52**	**27.23**
DINOv2	25.0%	64	Random	94.03	93.87	1.81	1.75
DINOv2	**12.5%**	**32**	**ECTS**	**63.13**	**58.63**	**32.72**	**36.99**
DINOv2	12.5%	32	Random	88.28	87.96	7.57	7.66
SigLIP	100.0%	196	Baseline	98.26	97.95	0.00	0.00
SigLIP	**75.0%**	**147**	**ECTS**	**95.85**	**95.48**	**2.41**	**2.47**
SigLIP	75.0%	147	Random	97.88	97.53	0.39	0.42
SigLIP	**50.0%**	**98**	**ECTS**	**89.19**	**89.26**	**9.07**	**8.69**
SigLIP	50.0%	98	Random	97.51	97.19	0.75	0.76
SigLIP	**25.0%**	**49**	**ECTS**	**75.48**	**76.06**	**22.78**	**21.88**
SigLIP	25.0%	49	Random	95.79	95.48	2.47	2.47
SigLIP	**12.5%**	**24**	**ECTS**	**68.24**	**69.03**	**30.02**	**28.92**
SigLIP	12.5%	24	Random	89.17	88.75	9.09	9.19

4.3 Selection-Strategy Ablation

Table 5 reports a full ablation across all three encoders on the complete 1,036-image test set. Relevance+magnitude scoring alone is markedly worse than random for DINOv2 (60.91% vs. 95.58%) and SigLIP (57.82% vs. 97.55%), but remains close to random for ViT (93.63% vs. 98.42%). The greedy diversity-aware selection step recovers substantial accuracy for all three encoders, but by very different margins (Table 6): +5.02pp (ViT), +22.88pp (DINOv2), +31.37pp (SigLIP), yet this recovery is insufficient to match random selection for DINOv2 and SigLIP.

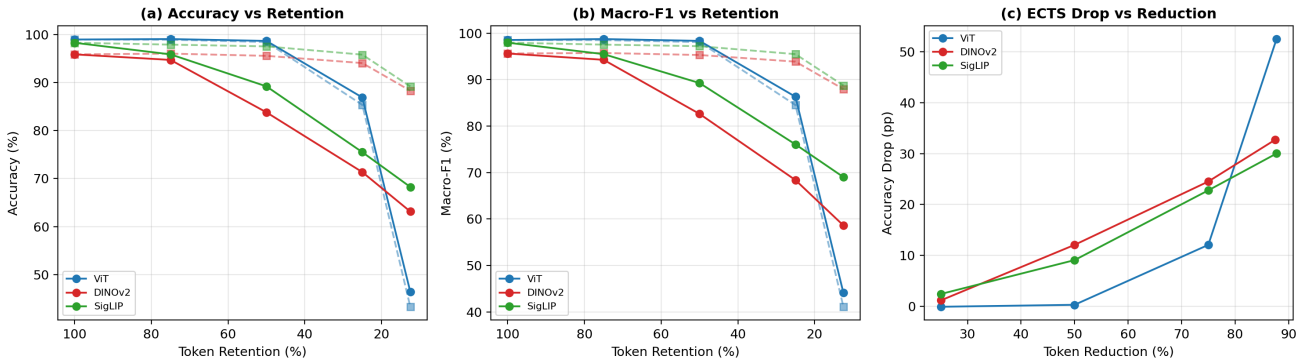


Figure 4: **Compression behavior across encoders.** (a) Accuracy and (b) Macro-F1 vs. token retention, ECTS (solid) vs. Random (dashed), all three encoders. (c) ECTS accuracy drop from baseline vs. token reduction percentage.

Table 4: One-sample t-test of ECTS accuracy against the random-seed distribution ($n = 5$), with Cohen's d effect size. Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

Model	Retention (%)	Random Mean	Random Std	ECTS	t	p	Cohen's d	Sig.
ViT	75.0	98.90	0.11	99.03	2.74	0.0517	1.23	ns
ViT	50.0	98.42	0.15	98.65	3.54	0.0241	1.58	*
ViT	25.0	85.33	0.55	86.87	6.33	0.0032	2.83	**
ViT	12.5	43.30	1.08	46.43	6.46	0.0030	2.89	**
DINOv2	75.0	96.00	0.20	94.69	-14.66	0.0001	-6.56	***
DINOv2	50.0	95.54	0.28	83.78	-95.11	0.0000	-42.53	***
DINOv2	25.0	94.03	0.65	71.33	-78.23	0.0000	-34.99	***
DINOv2	12.5	88.28	0.39	63.13	-144.34	0.0000	-64.55	***
SigLIP	75.0	97.88	0.14	95.85	-33.20	0.0000	-14.85	***
SigLIP	50.0	97.51	0.26	89.19	-71.83	0.0000	-32.12	***
SigLIP	25.0	95.79	0.38	75.48	-118.35	0.0000	-52.93	***
SigLIP	12.5	89.17	1.28	68.24	-36.47	0.0000	-16.31	***

Table 5: **Selection-strategy ablation** at 50% retention, full 1,036-image test set. **ECTS (full method)** rows in bold.

Encoder	Strategy	Acc. (%)	Std	F1 (%)	Std
ViT	Random	98.42	0.11	97.99	0.19
ViT	Relevance-only	87.55	0.00	86.74	0.00
ViT	Magnitude-only	95.85	0.00	95.23	0.00
ViT	Relevance+Magnitude	93.63	0.00	92.53	0.00
ViT	**ECTS (full method)**	**98.65**	**0.00**	**98.34**	**0.00**
DINOv2	Random	95.58	0.34	95.36	0.32
DINOv2	Relevance-only	61.29	0.00	59.61	0.00
DINOv2	Magnitude-only	66.02	0.00	66.49	0.00
DINOv2	Relevance+Magnitude	60.91	0.00	57.85	0.00
DINOv2	**ECTS (full method)**	**83.78**	**0.00**	**82.63**	**0.00**
SigLIP	Random	97.55	0.24	97.19	0.26
SigLIP	Relevance-only	58.11	0.00	58.14	0.00
SigLIP	Magnitude-only	72.20	0.00	72.14	0.00
SigLIP	Relevance+Magnitude	57.82	0.00	57.73	0.00
SigLIP	**ECTS (full method)**	**89.19**	**0.00**	**89.26**	**0.00**

Table 6: Accuracy gain from adding greedy diversity-aware selection to Relevance+Magnitude scoring alone, and remaining gap to Random, at 50% retention.

Encoder	Diversity Gain (pp)	ECTS−Random (pp)
ViT	5.02	0.23
DINOv2	22.88	-11.80
SigLIP	31.37	-8.36

4.4 Qualitative Token Selection Patterns

Figure 5 shows selected tokens at 50% retention. ECTS produces spatially contiguous, foreground-concentrated selections on ViT, nearly fully excluding plain background regions. DINOv2's selections are, in several cases, visually similar, comparably concentrated on foreground content, despite its far larger accuracy drop. SigLIP's selections are visibly less contiguous, with isolated background cells persisting alongside a foreground cluster. Two further examples are provided in Appendix A.

4.5 Inference Latency

Table 7 reports measured per-image latency. Token-count reduction does not translate into wall-clock speedup under our implementation: at 50% retention, ECTS inference is approximately 3× *slower* than the full-token baseline for all three encoders, due to the sequential, per-sample greedy selection loop. A net speedup is only achieved at 12.5% retention (1.10–1.14× across encoders).

Table 7: Measured per-image inference latency (batch size 1, RTX 3050, 200 timed iterations after warmup) and speedup relative to each model's own full-token (100%) baseline.

Model	Retention (%)	Tokens	Latency (ms)	Speedup
ViT	100.0	196	11.62 ± 0.17	1.00×
ViT	75.0	147	47.66 ± 2.13	0.24×
ViT	50.0	98	33.00 ± 1.06	0.35×
ViT	25.0	49	18.04 ± 0.42	0.64×
ViT	12.5	24	10.31 ± 0.22	1.13×
DINOv2	100.0	256	14.92 ± 0.25	1.00×
DINOv2	75.0	192	61.86 ± 2.26	0.24×
DINOv2	50.0	128	43.44 ± 2.58	0.34×
DINOv2	25.0	64	23.41 ± 2.09	0.64×
DINOv2	12.5	32	13.56 ± 0.90	1.10×
SigLIP	100.0	196	12.30 ± 0.18	1.00×
SigLIP	75.0	147	48.57 ± 3.04	0.25×
SigLIP	50.0	98	33.01 ± 1.47	0.37×
SigLIP	25.0	49	18.22 ± 0.64	0.68×
SigLIP	12.5	24	10.79 ± 0.85	1.14×

4.6 Representation Drift

Table 8 reports cosine similarity and relative L2 distance between full-token and ECTS-pruned pooled embeddings. ViT's pruned embedding remains close to its full-token embedding at 50% retention (mean cosine similarity 0.969, std 0.026). DINOv2 and SigLIP show substantially lower similarity (0.908, 0.888) and 3–7× higher variance (std 0.097, 0.173). The ordering holds at 25% retention (ViT 0.847/std 0.056; DINOv2 0.821/std 0.149;

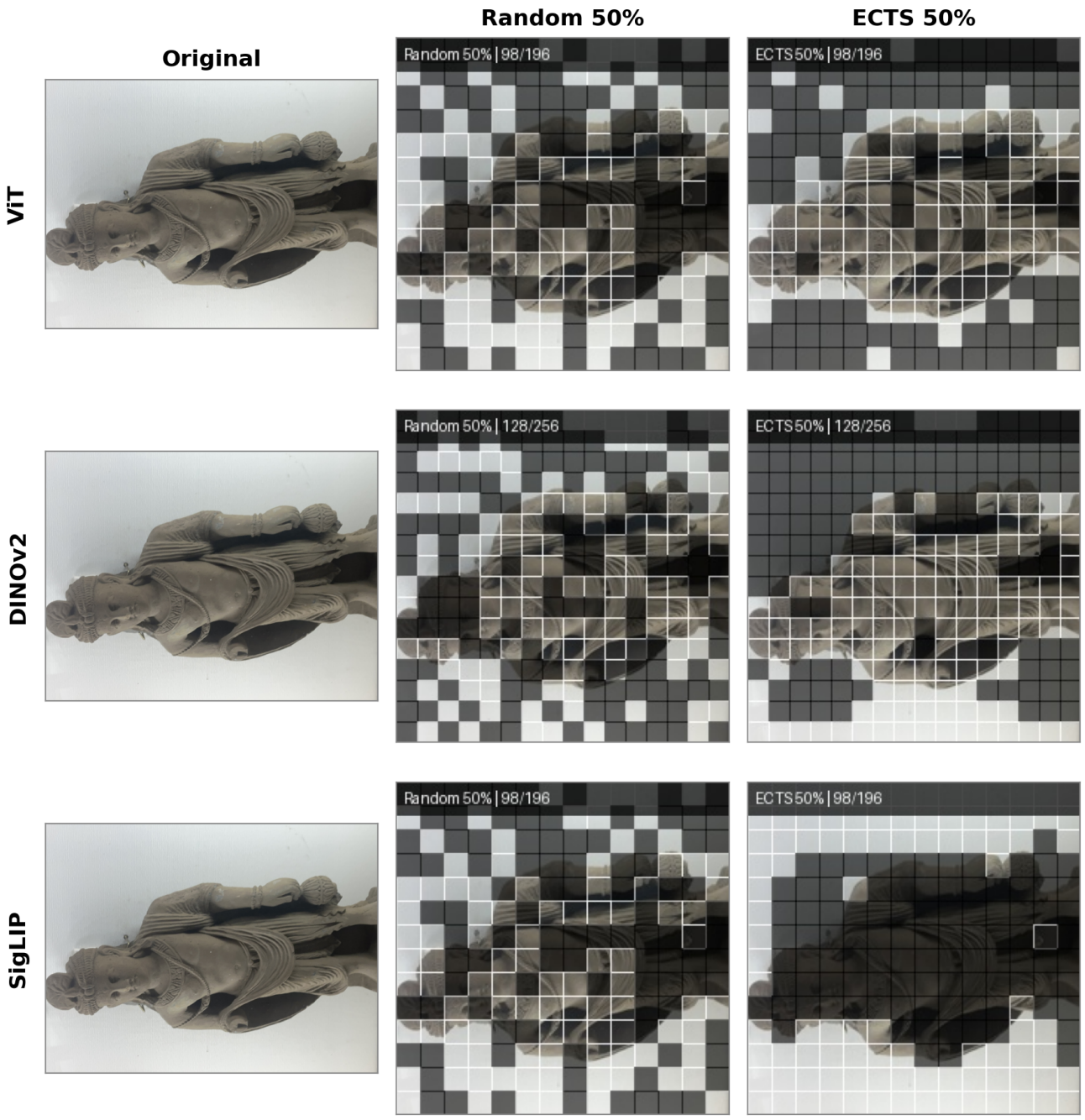


Figure 5: Qualitative token selection at 50% retention: Original, Random, and ECTS, across all three encoders, for a representative test image.

SigLIP 0.800/std 0.224), though the mean L2 drift magnitude converges across encoders at this more aggressive level: it is the *variance* of drift, not its mean magnitude, that consistently separates the encoders.

Table 8: Representation-drift diagnostic: cosine similarity and relative L2 distance between full-token and ECTS-pruned pooled embeddings, 300 test images, at 50% and 25% retention.

Model	Retention (%)	Cosine Sim. (mean ± std)	L2 Drift (mean ± std)
ViT	50	0.969 ± 0.026	0.235 ± 0.089
ViT	25	0.847 ± 0.056	0.541 ± 0.105
DINOv2	50	0.908 ± 0.097	0.378 ± 0.196
DINOv2	25	0.821 ± 0.149	0.542 ± 0.244
SigLIP	50	0.888 ± 0.173	0.361 ± 0.293
SigLIP	25	0.800 ± 0.224	0.514 ± 0.338

5 Discussion

5.1 Redundancy Is Not the Same as Selectability

The clearest finding of this study is that visual token redundancy and the ability of a lightweight, training-free heuristic to *locate* that redundancy are separate properties. DINOv2 and SigLIP both tolerate substantial random token removal, demonstrating that most of their tokens carry redundant information for this task. Yet the same relevance-and-magnitude scoring strategy that successfully isolates the informative subset in ViT actively identifies a *worse-than-random* subset in DINOv2 and SigLIP. The full ablation (Table 5) sharpens this further: relevance+magnitude scoring alone is not merely unhelpful for DINOv2/SigLIP, it is *actively harmful*, falling far below random. The diversity mechanism performs disproportionately more corrective work for these two encoders (Table 6), essentially rescuing the scoring criterion's poor choices, yet even this substantial rescue is insufficient to match trivial random selection.

This suggests the scoring criterion is implicitly tuned to properties of ViT's representation space, likely its CLS-token-centric, supervised-pretraining-influenced feature geometry, that do not hold for self-supervised (DINOv2) or contrastive (SigLIP) representations, where discriminative information may be distributed more evenly, or encoded differently, across spatial tokens.

5.2 Spatial Plausibility Does Not Guarantee Preserved Accuracy: A Representation-Drift Mechanism

The qualitative token maps (Figure 5) complicate a simple explanation: DINOv2's ECTS selection is, in several cases, spatially comparable to ViT's, yet its accuracy collapse is far worse. This motivated the representation-drift diagnostic (Table 8). ViT's pruned embedding remains close to its full-token embedding, with low variance across images. DINOv2 and SigLIP both show substantially lower similarity and, critically, 3–7$\times$ higher *variance*: their representations do not drift a uniformly larger amount under pruning, they drift *unpredictably*, sometimes remaining stable, sometimes shifting sharply. This pattern holds as a trend across both 50% and 25% retention, though the mean drift magnitude converges at 25%, indicating drift *instability*, not mean magnitude, is the robust discriminator.

We interpret this as evidence that the failure is not primarily about which spatial regions ECTS selects, but about how unevenly discriminative information is distributed across DINOv2/SigLIP's tokens relative to ViT, a property plausibly linked to their self-supervised and contrastive pretraining objectives, which do not use a single classification target to concentrate task-relevant information into an easily-identifiable token subset the way supervised ViT pretraining does. This is consistent with recent findings on anomalous "artifact tokens" and attention sinks within vision transformers [30], and echoes the visual under-conditioning failure mode identified by Venkatraman et al. [31] in self-evolving multimodal models, where insufficient or unreliable dependence on visual tokens similarly emerges as a representation-level, not purely architectural, phenomenon.

5.3 Theoretical Efficiency Gains Are Not Automatically Realized

Even where token reduction is accuracy-preserving (ViT at 50%), it does not automatically translate to faster inference (Table 7). The theoretical quadratic reduction in pairwise self-attention computation is real, but is outweighed in our implementation by the cost of the token-selection mechanism itself, a sequential, non-vectorized greedy search. This is an implementation property of ECTS as specified here, not an inherent property of encoder-aware compression; a batched or approximate top-k selection strategy would likely recover most of the theoretical speedup, in line with training-free methods that do report measured speedups through more efficient implementations [28, 29]. We report this result because claims of "efficient" compression should be evaluated on measured latency, not on token count alone.

5.4 Implications for Deployment-Oriented Museum Recognition Systems

These findings are relevant to systems in the spirit of Khan et al.'s AR-based Taxila Museum guide [1], which targets real-time, resource-constrained inference, and to broader museum management systems such as MAMS-Net [6] that build on encoders like ConvNeXt-V2 and DINOv2. Our results suggest that (1) encoder choice for such a system should account for both raw accuracy and compression amenability, since ViT-family encoders may be preferable specifically because they compress more predictably under simple heuristics, independent of their standalone accuracy, and (2) any proposed compression scheme intended for deployment must be validated with measured latency, since token-count-based savings are not guaranteed to hold in practice.

5.5 Limitations

This study evaluates a single dataset (TAD-8K, 40 classes) and a single compression method (ECTS); we do not claim the encoder-dependence pattern generalizes to all contextual selection strategies or visual domains. The latency benchmark reflects one implementation and one GPU (RTX 3050); relative overhead may differ with a vectorized selection implementation or different hardware. The qualitative token analysis is descriptive and based on a small sample of images; it motivates, but does not confirm, the feature-level hypothesis proposed to explain DINOv2/SigLIP's failure mode. ConvNeXt-V2 was included in the architecture comparison (Table 2) but excluded from the token compression analysis, as it is a convolutional architecture without a discrete token sequence or self-attention mechanism to which ECTS applies.

6 Conclusion

We show that a training-free, encoder-aware visual token compression method preserves classification accuracy under moderate compression for ViT-B/16 but underperforms random token removal for DINOv2-B/14 and SigLIP-B/16, despite both encoders exhibiting substantial token redundancy under random removal. A full ablation shows this is because the underlying scoring criterion is actively harmful, not merely uninformative, for these two encoders, and a representation-drift diagnostic links the failure to unstable, high-variance shifts in the pruned representation rather than poor spatial token selection. We further show that the token-count reduction achieved does not translate into measured latency improvement under our implementation until retention drops to 12.5%, underscoring the importance of reporting real inference time alongside token-count metrics in future compression work. Together, these results indicate that compression tolerance and selection reliability are distinct properties that must be evaluated separately per encoder, and that an honest account of a method's encoder-dependent limits is more useful to practitioners than a report scoped to its successes alone.

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Appendix

A Additional Qualitative Token Selection Examples

Figures 6 and 7 extend Figure 5 with two further representative test images, confirming the pattern reported in Section 4: ECTS selections on ViT and DINOv2 are often spatially comparable to one another, and both are more foreground-concentrated than SigLIP's, despite the substantial accuracy gap between ViT and the other two encoders under ECTS compression (Table 3).

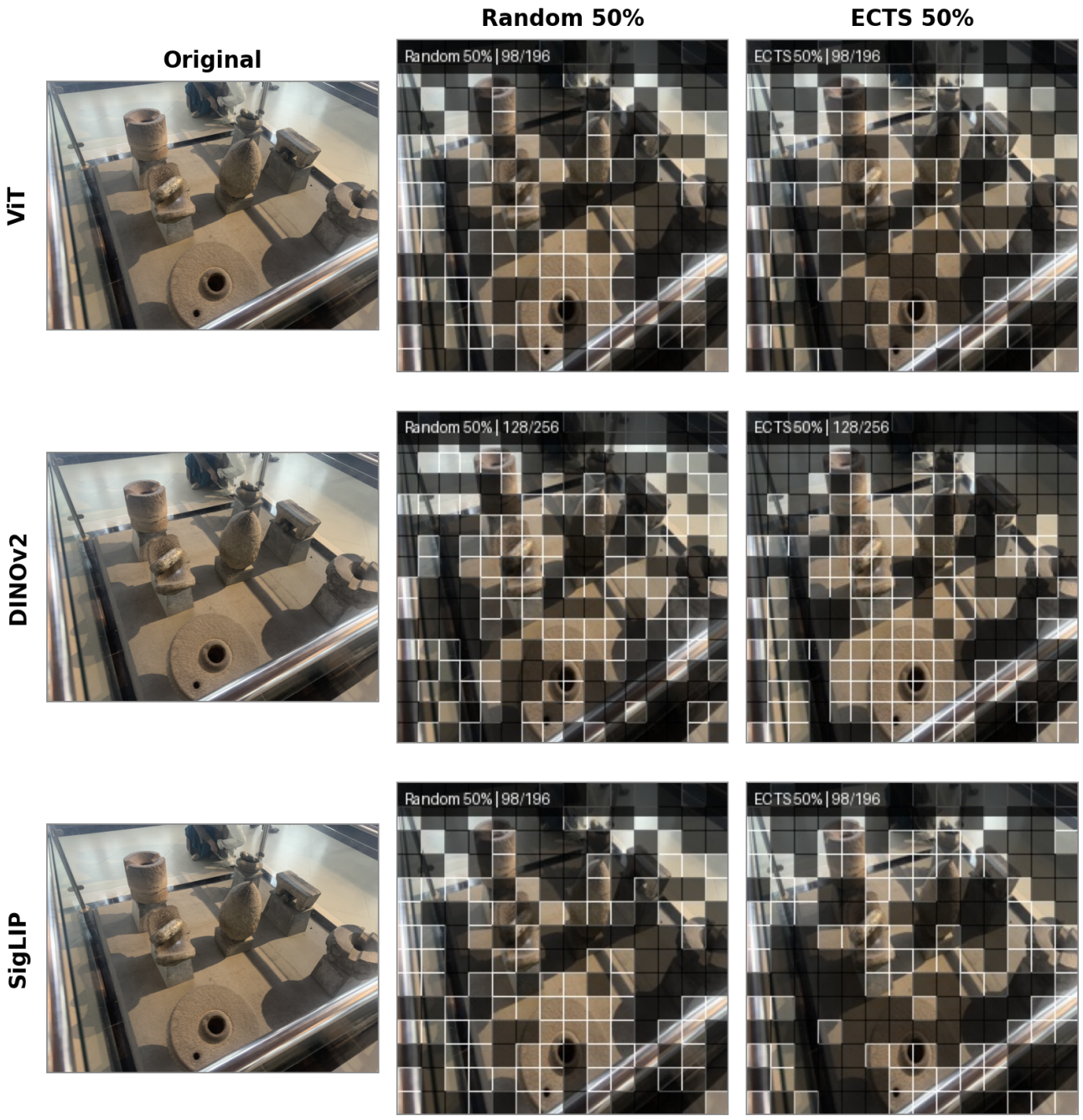


Figure 6: Qualitative token selection at 50% retention, additional example (museum display case).

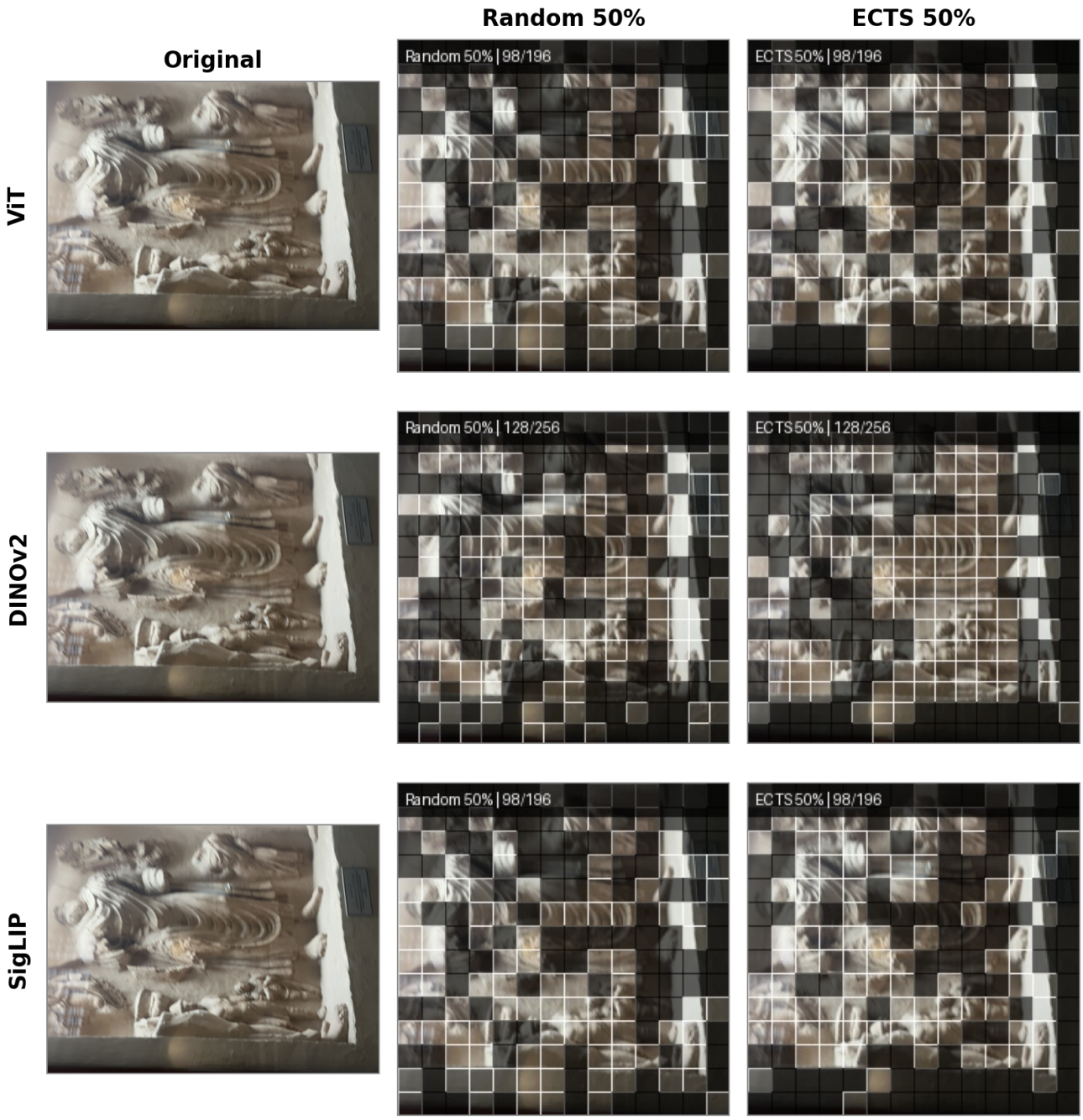


Figure 7: Qualitative token selection at 50% retention, additional example (relief carving, full-frame subject).

B Per-Class Classification Report

Table 9 reports per-class precision, recall, and F1-score for ConvNeXt-V2 Base (the best-performing encoder) on the TAD-8K test set. Of 40 classes, 32 achieve a perfect F1-score of 1.0000; the 8 classes listed below account for all residual test-set errors. The lowest-performing class (`artifact_13`) has only 7 test images, consistent with confusion driven primarily by small per-class sample size rather than a systematic visual ambiguity.

Table 9: Per-class precision, recall, and F1-score for ConvNeXt-V2 Base on the TAD-8K test set (40 classes, 1,036 images). Classes not listed reach 1.0000 across all three metrics.

Class	Precision	Recall	F1	Support
artifact_13	1.0000	0.7143	0.8333	7
artifact_14	0.8182	1.0000	0.9000	9
artifact_17	1.0000	0.9667	0.9831	30
artifact_20	1.0000	0.8824	0.9375	34
artifact_21	0.8333	1.0000	0.9091	30
artifact_28	0.9730	1.0000	0.9863	36
artifact_32	1.0000	0.9583	0.9787	24
artifact_34	1.0000	0.9630	0.9811	27
Macro avg	0.9906	0.9871	0.9877	1036
Weighted avg	0.9927	0.9913	0.9914	1036

C Training Curves

Figure 8 shows training and validation accuracy, loss, and Macro-F1 for ConvNeXt-V2 Base across training epochs, illustrating convergence behavior and the early-stopping point (best validation Macro-F1 of 99.53% at epoch 8, prior to the onset of overfitting visible after epoch 10).



Figure 8: Training and validation (a) accuracy, (b) loss, and (c) Macro-F1 vs. epoch, ConvNeXt-V2 Base.

D Prior Rejected Candidates: Full Results

Section 3 summarizes two rejected training-free candidates evaluated before ECTS. Table 10 reports their full results across all four retention levels, for completeness and reproducibility.

Table 10: Full results for rejected candidate methods: magnitude-based Top-K and ECRAS (magnitude + cosine-diversity redundancy scoring), compared against the Random baseline reported in the main text.

Model	Retention	Random (%)	Top-K (%)	ECRAS (%)
ViT	75%	98.90	98.63	98.94
ViT	50%	98.42	96.04	97.30
ViT	25%	85.33	74.03	78.67
ViT	12.5%	43.30	35.91	41.80
DINOv2	75%	96.00	88.80	91.51
DINOv2	50%	95.54	63.13	68.63
DINOv2	25%	94.03	23.75	35.71
DINOv2	12.5%	88.28	12.93	24.90
SigLIP	75%	97.88	92.76	93.63
SigLIP	50%	97.51	68.73	75.77
SigLIP	25%	95.79	30.79	43.73
SigLIP	12.5%	89.17	13.51	26.25

E Reproducibility Details

All experiments were run with Python 3.10.10, PyTorch 2.5.1+cu121, torchvision 0.20.1+cu121, and `timm` 1.0.24, on a single NVIDIA GeForce RTX 3050 (8GB). Random seeds for the 5-seed random token removal baseline were fixed at 42, 123, 456, 789, and 1001. All test-set inference used batch size 32 (batch size 1 for the latency benchmark specifically, to expose per-sample overhead), automatic mixed precision, and `num_workers=0`. A manual forward-pass equivalence check (saved predictions vs. fresh inference vs. block-by-block manual forward) confirmed zero prediction and zero maximum logit difference for all three transformer baselines prior to running any compression experiments.